TWO VARIABLES, ONE LINE

Linear Equations in Two Variables

✓ **Linear equation in two variables:**

- A **linear equation in two variables** (a letter that stands for an unknown number) is an equation that has two unknowns (usually x and y), each with power 1, and no term in which x and y are multiplied together.
- Examples: $2x + 3y = 12$, $y = 5x - 1$, $x - y = 0$. Not linear: $x^2 + y = 4$ (power 2), $xy = 6$ (x and y multiplied).

✓ **Standard form:**

- Every linear equation in two variables can be written as **ax + by + c = 0**.
- Here a, b and c are real numbers, and **a and b are not both zero** (if both were zero, there would be no variable left).

✓ **Coefficients and constant term:**

- **Coefficient:** the number multiplied with a variable. In ax + by + c = 0, a is the coefficient of x and b is the coefficient of y.
- **Constant term:** the number with no variable. In ax + by + c = 0, it is c.
- Both sides of an equation stand for the **same quantity**, written in two ways.

✓ **Fractions as coefficients:**

- a, b and c can be fractions. To make the equation neater, multiply the whole equation by the LCM of the denominators.
- Example: $(1/4)x + (3/5)y - 1/2 = 0$. LCM of 4, 5, 2 is 20. Multiplying by 20 gives **5x + 12y − 10 = 0**.

✓ **Equations with only one variable:**

- An equation like ax + c = 0 also counts as a linear equation in two variables, with b = 0.
- Example: 7 + 4x = 0 is written as **4x + 0 × y + 7 = 0**, so a = 4, b = 0, c = 7.

✓ **Same equation, different forms:**

- Multiplying or dividing the whole equation by any non-zero number gives the **same** equation in another form.
- Example: 5x + 2y − 2.7 = 0 and −5x − 2y + 2.7 = 0 are the same equation (the second is the first multiplied by −1). So the values of a, b, c are not fixed for one equation.

✓ **Worked examples:**



Fig. 1: Radha at the fruit cart: mangoes ₹60/kg, bananas ₹50/kg

- **Example 1 (forming an equation):** Radha buys some kilograms of mangoes at ₹60 per kg and some

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kilograms of bananas at ₹50 per kg, and pays ₹280 in all.

- Let x = kg of mangoes and y = kg of bananas.
- Then cost of mangoes = ₹60x and cost of bananas = ₹50y, so **60x + 50y = 280**. Both sides stand for the total cost in ₹.

- **Example 2 (standard form):** write each equation as ax + by + c = 0 and find a, b, c.

Equation	Standard form	a	b	c
5x + 2y = 2.7	5x + 2y − 2.7 = 0	5	2	−2.7
x/2 − 4 = 3y/2	(1/2)x − (3/2)y − 4 = 0	1/2	−3/2	−4
4 = 1.5x − 1.3y	1.5x − 1.3y − 4 = 0	1.5	−1.3	−4
3y = 5	0 × x + 3y − 5 = 0	0	3	−5

✓ **Solved questions:**

Q1. Write a linear equation in two variables in which a = 3, b = 0 and c = −1/5.

➢ 3x + 0 × y − 1/5 = 0, that is, **3x − 1/5 = 0** (or 15x − 1 = 0).

Q2. Complete the table after writing the given equations in standard form: y − 15 = √2 x; 3y − 2x = 0; 5x = 3y; x = 8; 3y = 1.

Equation	Standard form	Coefficient of x	Coefficient of y	Constant
y − 15 = √2 x	√2 x − y + 15 = 0	√2	−1	15
3y − 2x = 0	−2x + 3y + 0 = 0	−2	3	0
5x = 3y	5x − 3y + 0 = 0	5	−3	0
x = 8	x + 0 × y − 8 = 0	1	0	−8
3y = 1	0 × x + 3y − 1 = 0	0	3	−1

Q3. (i) The cost of a notebook is twice the cost of a pen. Taking the cost of a notebook as ₹t and of a pen as ₹p, Charlie wrote t = 2p, whereas Meera wrote p = 2t. Which is correct? (ii) Two Indian batsmen together scored 176 runs in a one-day match. Manisha wrote x + y = 176, where x and y are their runs. Is this correct?

➢ (i) **Charlie (t = 2p)** is correct.

➢ The notebook costs twice the pen, so t is 2 times p.

➢ Check: if a pen costs ₹10, a notebook costs ₹20, and t = 2p gives 20 = 2 × 10.

➢ (ii) **Yes.** The runs of the two batsmen add up to 176, so x + y = 176 is correct.

Solution of a Linear Equation in Two Variables

✓ **Solution of an equation:**

- A **solution** of a linear equation in two variables is a pair of values, one for x and one for y, that makes the left side equal to the right side.
- When these values make the two sides equal, we say they **satisfy** (make both sides of the equation equal) the equation.

✓ **Ordered pair:**

- A solution is written as **(x, y)**, with the x-value first and the y-value second. This is an **ordered pair**.

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- The order matters: (2, 3) means x = 2, y = 3, while (3, 2) means x = 3, y = 2. They are different points.

✓ **Number of solutions:**

- A linear equation in two variables has **infinitely many solutions**.
- How to get them: give x any value; the equation then has only y left, which gives one value of y. Each value of x gives a new solution, and x can take endless values.

✓ **Graph of the equation:**

- All the solutions, when plotted, lie on **one straight line**. This line is the **graph** of the equation.
- Every point on the line is a solution, and every solution is a point on the line. A point not on the line is not a solution.
- Rearranging an equation does not change its graph. Example: 3x + 2y = 12 and y = −(3/2)x + 6 are the same line.

✓ **Quick way to draw the graph:**

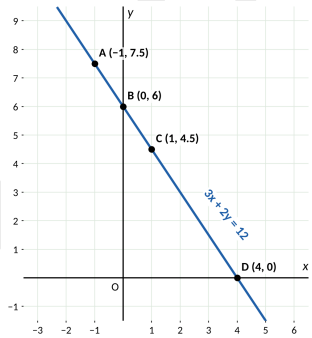
1) Put **x = 0** and find y. This gives the point where the line cuts the **y-axis**.
2) Put **y = 0** and find x. This gives the point where the line cuts the **x-axis**.
3) Plot both points and join them with a straight line. Plot a third point to check.

✓ **Line through the origin:**

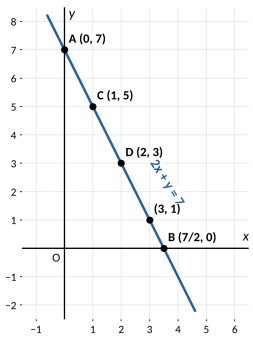
- If **c = 0**, the equation is ax + by = 0. Then x = 0, y = 0 satisfies it, so the line always passes through the **origin (0, 0)**.
- Special cases: x = 0 is the **y-axis**, and y = 0 is the **x-axis**.
- For such a line, x = 0 and y = 0 both give the same point (0, 0). So for a second point, put another value, such as x = 1.

✓ **Worked examples:**

- **Example 1:** for 3x + 2y = 12: (2, 3) is a solution, as 3 × 2 + 2 × 3 = 12; (4, 0) is a solution, as 12 + 0 = 12; (1, 3) is not, as 3 + 6 = 9 ≠ 12.
- **Example 2:** checking more points for 3x + 2y = 12 (graph (a) below): (1, 9/2) gives 3 + 9 = 12 ✓; (0, 6) gives 0 + 12 = 12 ✓; (4, 1) gives 12 + 2 = 14 ✗; (1/2, 21/4) gives 3/2 + 21/2 = 12 ✓. Other solutions: (−1, 7.5), (1, 4.5), (6, −3), and so on, without end.
- **Example 3:** find two solutions of 2x + y = 7, draw its graph (graph (b) below) and find 4 more solutions.
  - x = 0 gives y = 7, so (0, 7); y = 0 gives 2x = 7, x = 7/2, so (7/2, 0).
  - From the graph, (1, 5), (2, 3), (3, 1) and (3/2, 4) are also solutions.



(a) 3x + 2y = 12



(b) 2x + y = 7

Fig. 2: Graphs of 3x + 2y = 12 and 2x + y = 7: every solution lies on the line

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- **Example 4:** draw the graphs of (i) 4x + 3y = 12: x = 0 gives (0, 4) and y = 0 gives (3, 0); (ii) 2x + 5y = 0: x = 0 gives (0, 0); x = 1 gives y = −2/5; x = 5 gives y = −2. The points (0, 0), (1, −2/5) and (5, −2) lie on a line through the origin.

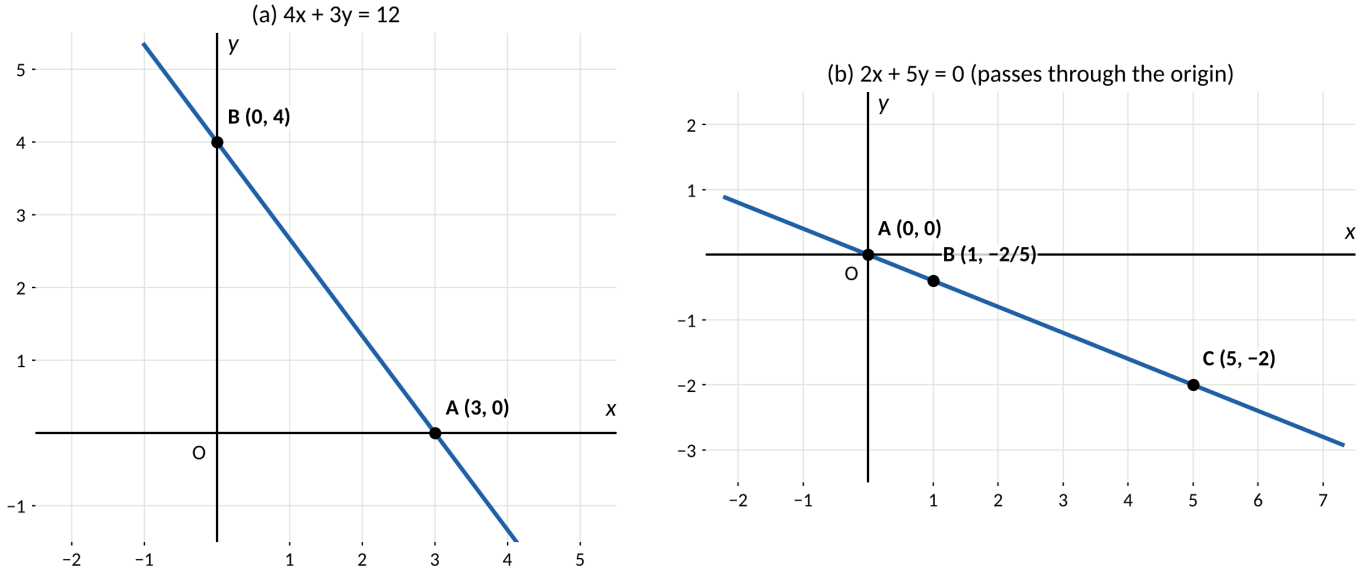


Fig. 3: Graphs of (a) 4x + 3y = 12 and (b) 2x + 5y = 0

- **Example 5:** (−3, 4) is a solution of both 3ax + 4y = −2 and 2x + by = 14. Find a and b.
  - Put x = −3, y = 4 in the first: −9a + 16 = −2, so −9a = −18 and **a = 2**.
  - Put x = −3, y = 4 in the second: −6 + 4b = 14, so 4b = 20 and **b = 5**.

✓ **Solved questions:**

Q1. Check if the ordered pair (4, 3) is a solution of 5x − 6y = 2. Explain.

➢ LHS = 5 × 4 − 6 × 3 = 20 − 18 = 2 = RHS. So **yes**, (4, 3) is a solution.

Q2. Find any two solutions of (i) 7x − 3y = 21 (ii) 2x + 3y = 5.

➢ (i) x = 0 gives y = −7; y = 0 gives x = 3.

➢ Solutions: **(0, −7), (3, 0)**.

➢ (ii) x = 1 gives 3y = 3, y = 1; x = 4 gives 3y = −3, y = −1.

➢ Solutions: **(1, 1), (4, −1)**.

Q3. m and n are unknown constants in 2mx + 3y = 7 and 4x + ny = −10. If (2, −1) is a solution of both, find m and n.

➢ First: 2m × 2 + 3 × (−1) = 7, so 4m = 10 and **m = 5/2**.

➢ Second: 4 × 2 + n × (−1) = −10, so −n = −18 and **n = 18**.

Q4. Find two solutions lying in different quadrants for each equation, and name the quadrants: (i) 5x + 3y = 7 (ii) 5x − 3y = 7 (iii) −5x + 3y = 7 (iv) −5x − 3y = 7.

➢ Quadrants: I (x +, y +), II (x −, y +), III (x −, y −), IV (x +, y −).

➢ (i) (2, −1) in **IV**; (−1, 4) in **II**.

➢ (ii) (2, 1) in **I**; (−1, −4) in **III**.

➢ (iii) (1, 4) in **I**; (−2, −1) in **III**.

➢ (iv) (−2, 1) in **II**; (1, −4) in **IV**.

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- Each pair checks out on substitution, and the points lie on the line when drawn on graph paper.

Q5. For the graph of 3x − 7y = 21 shown, does the point C (2, 3) lie on the line? Does it satisfy the equation? Can points that do not lie on the line satisfy the equation?

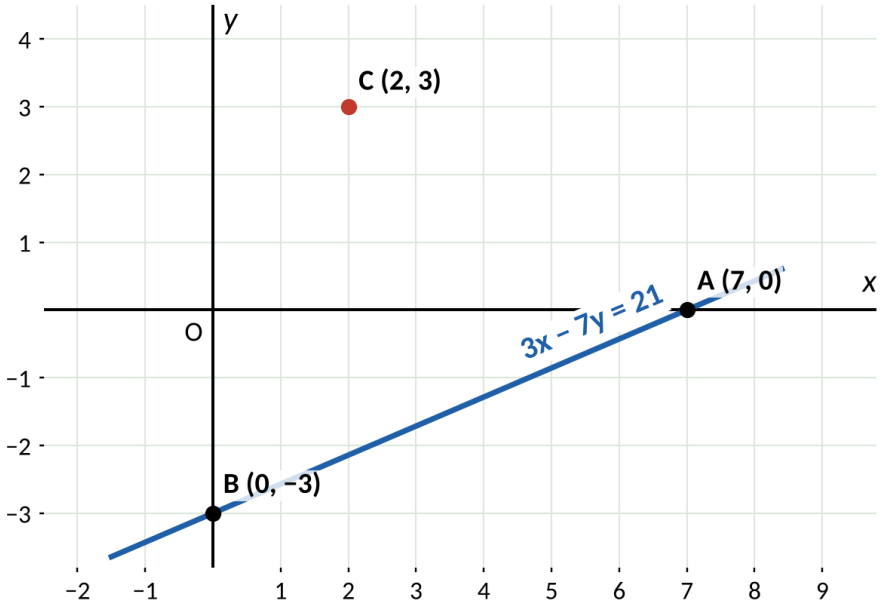


Fig. 4: Graph of 3x − 7y = 21 and the point C (2, 3)

- **No**, C does not lie on the line.
- 3 × 2 − 7 × 3 = 6 − 21 = −15 ≠ 21, so C does **not** satisfy the equation.
- **No.** The line is made of exactly those points that satisfy the equation, so a point off the line can never satisfy it.

Q6. True or False? Justify. (i) A linear equation in two variables has only one solution. (ii) Its graph always passes through the origin. (iii) It can never have rational solutions. (iv) x = 3 is a valid linear equation in two variables. (v) 2x + 3y = 7 has infinitely many solutions. (vi) (1, 2) is a solution of 2x + 3y = 7.

- (i) **False**: it has infinitely many solutions.
- (ii) **False**: only when c = 0.
- Example: x + y = 1 does not pass through (0, 0).
- (iii) **False**: (2, 1) is a rational solution of 2x + 3y = 7.
- (iv) **True**: x = 3 is x + 0 × y − 3 = 0, with a = 1, b = 0.
- (v) **True**: every value of x gives a matching y.
- (vi) **False**: 2 × 1 + 3 × 2 = 8 ≠ 7.

Q7. (i) Show that 3x + 4y = 7 and 6x + 8y = 14 have the same set of solutions. (ii) Show that ax + by = c and kax + kby = kc (k ≠ 0) have the same set of solutions.

- (i) 6x + 8y = 14 is 2 × (3x + 4y) = 2 × 7.
- If (x, y) satisfies the first, doubling both sides shows it satisfies the second; if it satisfies the second, halving both sides shows it satisfies the first.
- So both have the same solutions (and the same line).
- (ii) If ax + by = c, multiplying by k gives kax + kby = kc.
- If kax + kby = kc, dividing by k (allowed, as k ≠ 0) gives ax + by = c.

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➢ So every solution of one is a solution of the other.

Slope of a Line

✓ **Slope:**

- The **slope** (or **gradient**) of a line tells how steep the line is.
- A steep hill is hard to climb or cycle up; a flat road is easy. In the same way, a steeper line has a bigger slope (in size).

✓ **Positive and negative slope:**

- **Positive slope:** the line goes **up** as we move from left to right.
- **Negative slope:** the line goes **down** as we move from left to right.

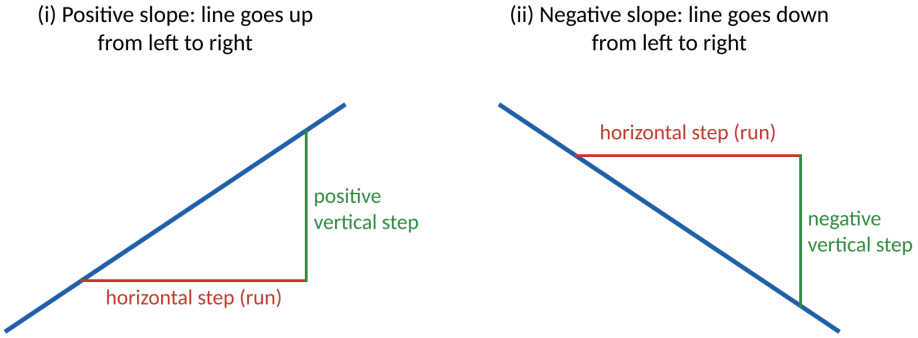


Fig. 5: Positive and negative slope

✓ **Rise and run:**

- Take two points A (x_1, y_1) and B (x_2, y_2) on the line.
- **Run:** the horizontal distance from A to B, $x_2 - x_1$. It is positive when we move right and negative when we move left.
- **Rise:** the vertical distance from A to B, $y_2 - y_1$. It is positive when we go up and negative when we go down.

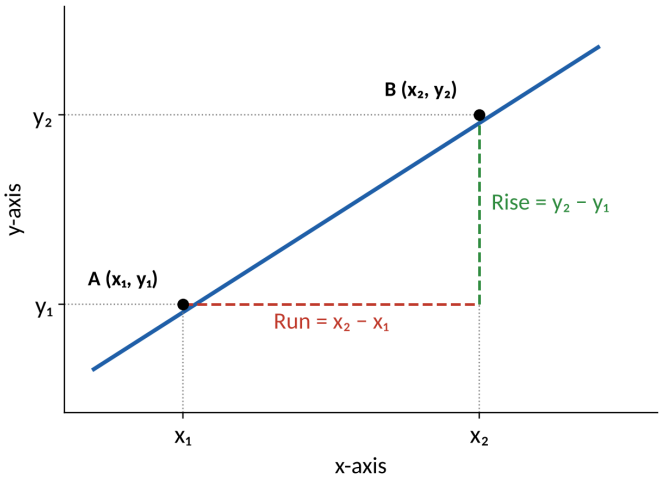


Fig. 6: Rise and run between two points on a line

✓ **Formula for slope:**

- **Slope = Rise ÷ Run = $(y_2 - y_1) \div (x_2 - x_1)$**

- Any two points on the line give the **same** slope.
- The order of the points does not matter: taking B first changes the sign of both rise and run, so the answer stays the same.

✓ **Slope-intercept form:**

- A line written as **y = mx + d** is in **slope-intercept form**.
- **m** is the slope.
- **d** is the **y-intercept**: the line cuts the y-axis at (0, d).

✓ **Proof that m is the slope:**

- Let A (x_1, y_1) and B (x_2, y_2) be on y = mx + d. So $y_1 = mx_1 + d$ and $y_2 = mx_2 + d$.
- Slope = $(y_2 - y_1) \div (x_2 - x_1) = [(mx_2 + d) - (mx_1 + d)] \div (x_2 - x_1) = m(x_2 - x_1) \div (x_2 - x_1) =$ **m**.
- The answer does not depend on which two points we took. This shows the slope is the same all along the line.

✓ **Slope from the standard form:**

- ax + by + c = 0 gives by = −ax − c, so **y = −(a/b)x − c/b** (when b ≠ 0).
- So, **slope m = −a/b** and **y-intercept d = −c/b**.

✓ **x-intercept:**

- The **x-intercept** is where the line cuts the x-axis. There, y = 0.
- For y = mx + d: 0 = mx + d, so **x = −d/m** (when m ≠ 0).

✓ **Four kinds of slope:**

- **m > 0:** line rises from left to right; as x increases, y increases. In y = 2x + 1, each step of 1 to the right goes 2 up.
- **m < 0:** line falls from left to right; as x increases, y decreases. In y = −3x + 4, each step of 1 to the right goes 3 down.
- **m = 0:** line is **horizontal** (parallel to the x-axis); y does not change. Example: y = 5.
- **Undefined slope:** line is **vertical** (parallel to the y-axis); x does not change. Example: x = 2. The run is 0, and dividing by 0 is not possible, so the slope is not defined. Such a line cannot be written as y = mx + d.

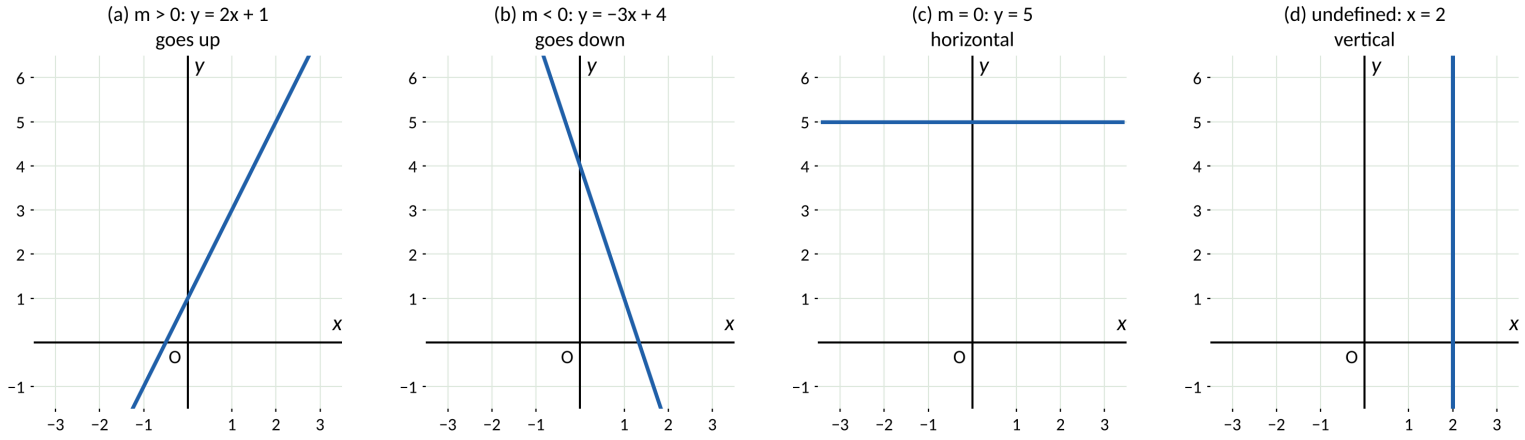


Fig. 7: Lines with positive, negative, zero and undefined slope

✓ **Worked examples:**

- **Example 1:** a line passes through A (2, 0) and B (4, 3)

- From A to B: run = 2, rise = 3, so slope = 3/2.
- Extending to D (8, 9): from B to D, run = 4 and rise = 6, so slope = 6/4 = 3/2 again.
- The slope is the same everywhere on the line.

- On this line, the point with x = 8 is **(8, 9)**, and the point with y = 6 is **(6, 6)** (each step of 2 to the right gives 3 up).

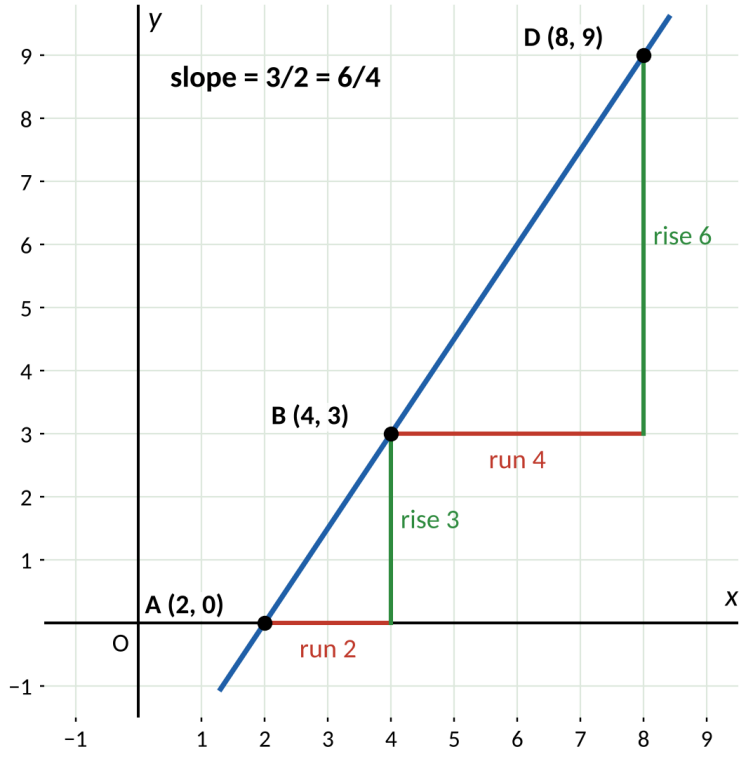


Fig. 8: The slope of a line is the same wherever it is measured

- **Example 2:** the line through A (−1, 0), B (1, 2) and C (4, 5): taking B and C, slope = (5 − 2)/(4 − 1) = 3/3 = **1**. Any other pair gives the same value.
- **Example 3:** the line through A (−4, 5) and B (−1, 2): slope = (2 − 5)/(−1 − (−4)) = −3/3 = **−1**. Taking B first: (5 − 2)/(−4 − (−1)) = 3/(−3) = −1. The order of the points does not matter.

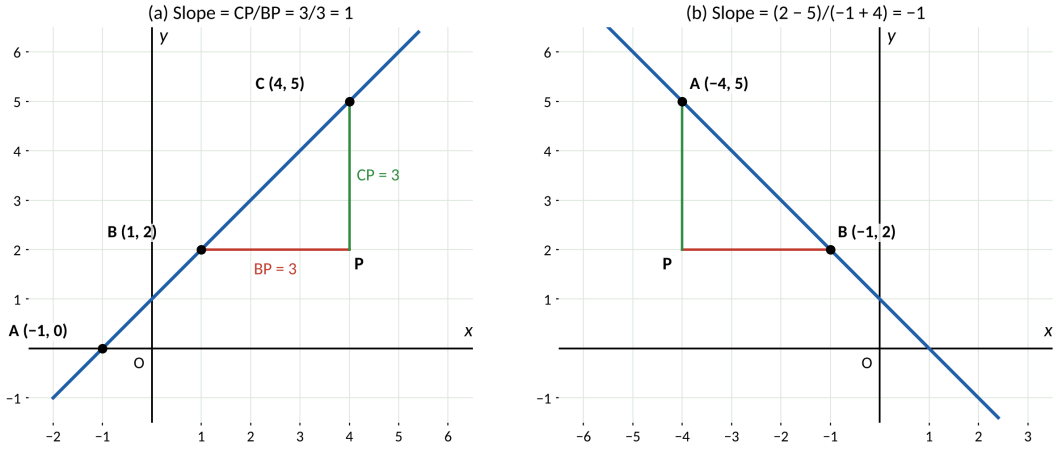


Fig. 9: Finding the slope of a line from two of its points

- **Example 4:** 5x + y = 3 is y = −5x + 3.
  - So m = −5: for every 1 unit increase in x, y decreases by 5. d = 3: the line cuts the y-axis at (0, 3)
  - Points: A (0, 3), B (1, −2), C (−1, 8)

o Check: A to B gives (−2 − 3)/1 = −5, and C to A gives (3 − 8)/(0 + 1) = −5.

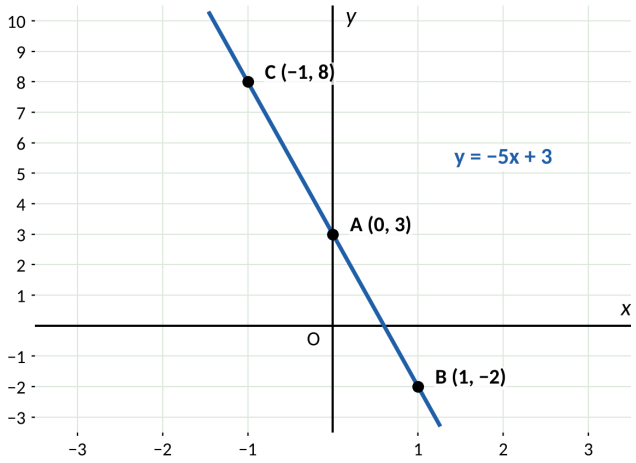


Fig. 10: Graph of y = −5x + 3

✓ **Solved questions:**

Q1. Three ski hills are shown: A (rise 60 m, run 70 m), B (rise 60 m, run 110 m) and C (rise 80 m, run 100 m). Rank them by steepness, from least to greatest.

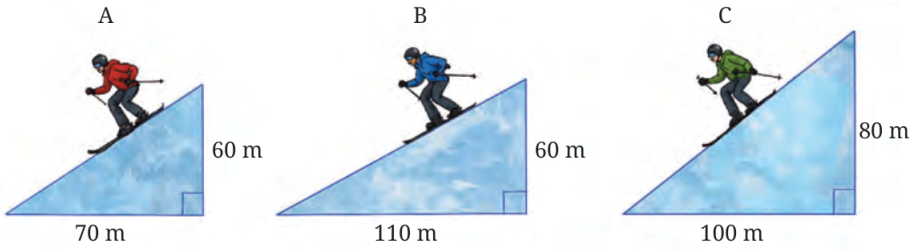


Fig. 11: Ski hills A, B and C

➢ Slope of A = 60/70 ≈ 0.86; slope of B = 60/110 ≈ 0.55; slope of C = 80/100 = 0.80.

➢ From least to greatest steepness: **B, C, A**.

Q2. The ramp at a loading dock rises 2.5 m over a run of 4 m. Find its slope.

➢ Slope = rise/run = 2.5/4 = **0.625** (= 5/8).

Q3. Find the slope of the line l in each diagram.

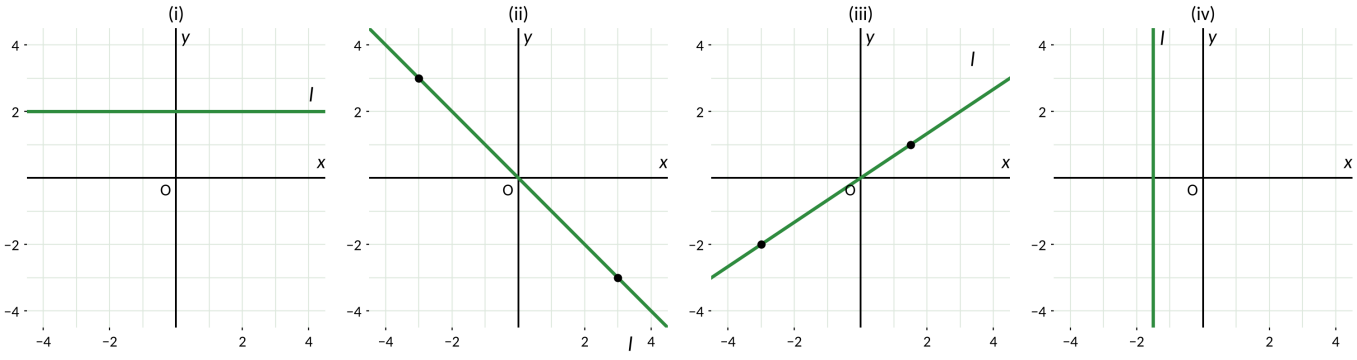


Fig. 12: Lines for Q3 (each small square = 1 unit)

➢ (i) l is horizontal: slope = **0**.

➢ (ii) l passes through (−3, 3) and (3, −3): slope = (−3 − 3)/(3 + 3) = **−1**.

➢ (iii) l passes through (−3, −2) and (1.5, 1): slope = 3/4.5 = **2/3**.

➢ (iv) l is vertical: slope is **undefined**.

Q4. The guideline for a wheelchair ramp is a rise of 1 cm for every 12 cm of horizontal length. What should be the horizontal length of the ramp if the stairs are 18 cm high?

➢ Slope = 1/12 = 18/(horizontal length), so horizontal length = 18 × 12 = **216 cm**.

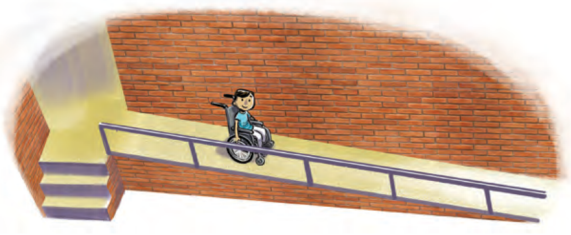


Fig. 13: A wheelchair ramp beside a staircase

Q5. The graph of y = 3x passes through (0, 0) and (2, 6). (i) Find its slope from the two points. (ii) What is its y-intercept and what does it mean? (iii) A water tank is filled so that the water level y (cm) after x minutes follows this graph. (a) How high is the water after 5 minutes? (b) After how many minutes is it 21 cm? (iv) Does (4, 10) lie on the line? (v) Where does it meet the line x = 3?

➢ (i) Slope = (6 − 0)/(2 − 0) = **3**.

➢ (ii) y-intercept = **0**: the line passes through the origin; the tank was empty at the start (level 0 at time 0).

➢ (iii)(a) y = 3 × 5 = **15 cm**. (b) 21 = 3x, so x = **7 minutes**.

➢ (iv) For x = 4, y = 12, not 10.

➢ So (4, 10) is **not** on the line.

➢ (v) Put x = 3 in y = 3x: y = 9.

➢ The lines meet at **(3, 9)**.

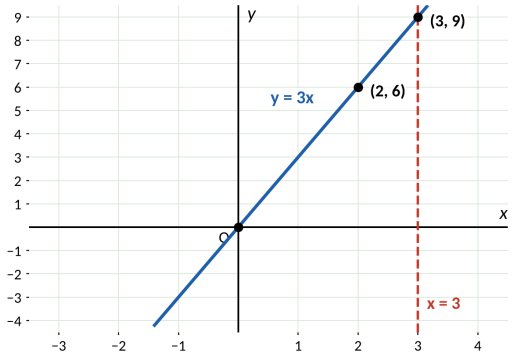


Fig. 14: Graph of y = 3x and the line x = 3

Q6. The equation F = (9/5)C + 32 changes Celsius to Fahrenheit. (i) Draw its graph (C on the x-axis, F on the y-axis). (ii) Change 30 °C to °F. (iii) Change 95 °F to °C. (iv) What is 0 °C in °F, and 0 °F in °C? (v) Is there a temperature with the same number on both scales?

➢ (i) The graph is a straight line with slope 9/5 and F-intercept 32.

➢ Plot (0, 32) and (100, 212) and join them (see the figure below).

➢ (ii) F = (9/5) × 30 + 32 = 54 + 32 = **86 °F**.

➢ (iii) 95 = (9/5)C + 32, so (9/5)C = 63 and C = **35 °C**.

➢ (iv) 0 °C gives F = **32 °F**. 0 °F gives (9/5)C = −32, so C = −160/9 ≈ **−17.8 °C**.

➢ (v) Put F = C: C = (9/5)C + 32, so −(4/5)C = 32 and C = **−40**. So **−40 °C = −40 °F**.

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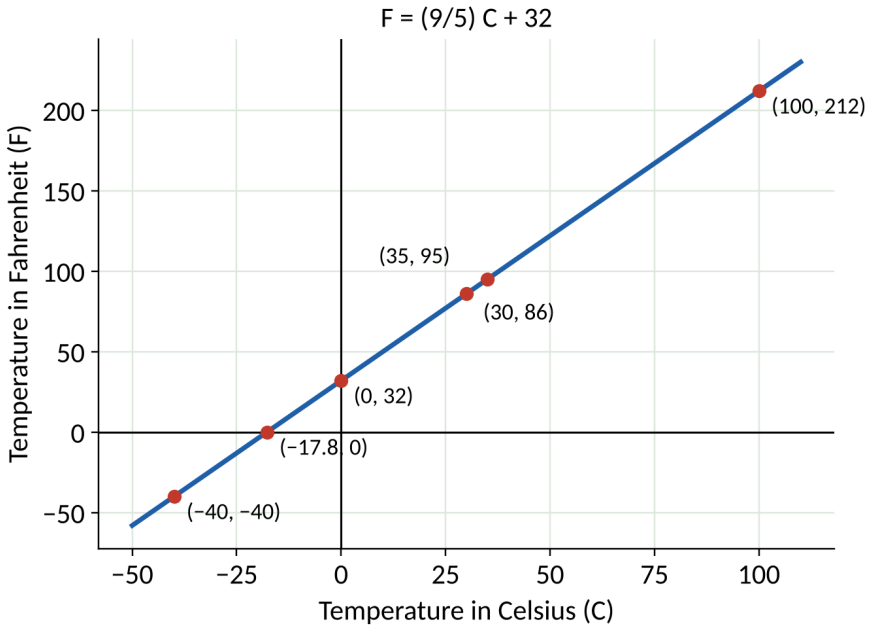


Fig. 15: Graph of F = (9/5)C + 32

Q7. Give a formula for the x-intercept of the line y = mx + c.

- At the x-axis, y = 0: 0 = mx + c, so **x = −c/m** (m ≠ 0)
- The line cuts the x-axis at (−c/m, 0).

Q8. How many lines (i) have a given slope? (ii) have a given slope and pass through a given point?

- (i) **Infinitely many**: all lines y = mx + d with the same m and any d.
- They are all parallel.
- (ii) **Exactly one**: among those parallel lines, only one passes through the given point.

Pair of Linear Equations in Two Variables

✓ **Pair of linear equations:**

- A **pair of linear equations** means two linear equations in the same two variables, taken together.
- General form: $\mathbf{a_1x + b_1y + c_1 = 0}$ and $\mathbf{a_2x + b_2y + c_2 = 0}$, where a_1, b_1 are not both zero and a_2, b_2 are not both zero.

✓ **Solution of a pair:**

- A **solution of a pair** is an ordered pair (x, y) that satisfies **both** equations at the same time.
- One equation alone has endless solutions; together, the two equations usually pin down just one.

✓ **Forming a pair from a situation:**

1) Choose letters for the two unknowns (say x and y) and write what each stands for.
2) Turn each piece of information into one equation.
3) Check that both equations use the same units.

✓ **Worked examples:**

- **Example 1:** two numbers have sum 25 and difference 11: **x + y = 25** and **x − y = 11**. Trying pairs (10, 15), (20, 5), (18, 7) shows (18, 7) works, so the numbers are 18 and 7.
- **Example 2:** at a science exhibition, 2 adult and 3 child tickets cost ₹600, and 3 adult and 2 child

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tickets cost ₹700.

- With ₹x per adult and ₹y per child: **2x + 3y = 600** and **3x + 2y = 700**. Adding: 5x + 5y = 1300, so x + y = 260; subtracting: x − y = 100.
- So x = 180 and y = 80: an adult ticket costs **₹180** and a child ticket **₹80**.

- **Example 3:** 1 paratha and 2 bowls of dahi cost ₹90; 2 parathas and 7 bowls of dahi cost ₹260.
  - With paratha ₹x and dahi ₹y: x + 2y = 90 and 2x + 7y = 260.
  - Doubling the first: 2x + 4y = 180; subtracting from the second: 3y = 80, so y = 80/3 ≈ ₹26.67 and x = 110/3 ≈ ₹36.67.



Fig. 16: Paratha and dahi prices

✓ **Solved questions:**

Form a pair of linear equations in two variables for each situation.

Q1. Let one movie ticket cost ₹x and one snack box ₹y. On day 1, a family buys 2 tickets and 3 snack boxes for ₹850; on day 2, 4 tickets and 1 snack box for ₹1100.

➢ **2x + 3y = 850** and **4x + y = 1100**.

Q2. Let the fixed taxi charge be ₹x and the charge per km ₹y. Sahil travelled 6 km and paid ₹122; Meena travelled 8 km and paid ₹160.

➢ **x + 6y = 122** and **x + 8y = 160**.

Q3. At a sports meet, adult tickets cost ₹150 and child tickets ₹100. 200 people attended and ₹25,000 was collected. Let x adult and y child tickets be sold.

➢ **x + y = 200** and **150x + 100y = 25000** (or 3x + 2y = 500).

Algebraic Methods of Solving a Pair of Linear Equations

✓ **Main idea:**

- Both algebraic methods turn the pair into **one equation with one variable**, which is easy to solve.
- The two methods are **substitution** and **elimination**.

✓ **Substitution method:**

- **Substitution method:** find one variable from one equation and put it into the other equation.
- **Steps:**

1) From either equation, write one variable in terms of the other (choose the equation where this is easiest).
2) Put (substitute) this expression in the **other** equation. Now only one variable is left; solve for it.

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3) Put this value back to get the second variable.
4) Check the answer in both original equations.

- It does not matter which variable is found first; the solution comes out the same.

Worked examples:

- **Example 1:** solve 7x − 15y = 2 … (1) and x + 2y = 3 … (2).
  - From (2): x = 3 − 2y … (3).
  - Put in (1): 7(3 − 2y) − 15y = 2, so 21 − 29y = 2, −29y = −19 and **y = 19/29**.
  - From (3): x = 3 − 38/29 = **49/29**. Check in (2): 49/29 + 38/29 = 87/29 = 3 ✓.
- **Example 2:** two angles of a triangle are x° and y° with y = 4x, and the third angle is 50°.
  - Angle sum: x + y + 50 = 180.
  - Putting y = 4x: 5x = 130, so **x = 26°** and **y = 104°**.
- **Example 3 (shape puzzle):** each shape stands for a number; the numbers at the ends are the sums of the rows and columns.

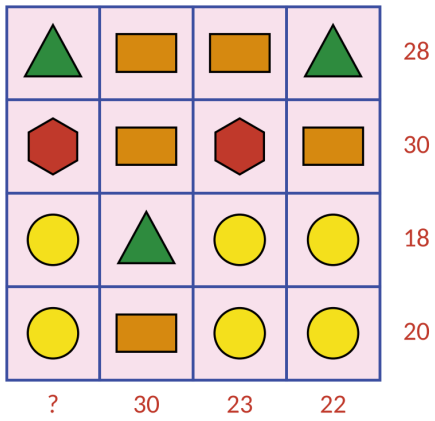


Fig. 17: Shape puzzle: find the value of each shape

  - Row 1: 2T + 2R = 28, so T + R = 14. Column 4: T + R + 2C = 22, so 14 + 2C = 22 and **C = 4**.
  - Row 3: 3C + T = 18, so **T = 6**; then **R = 8**. Row 2: 2H + 2R = 30, so **H = 7**.
  - Check: column 2: 3R + T = 24 + 6 = 30 ✓; column 3: R + H + 2C = 8 + 7 + 8 = 23 ✓. The missing column 1 total = T + H + 2C = 6 + 7 + 8 = **21**.

✓ **Elimination method:**

Elimination method: make the coefficients (the numbers multiplied with the variables) of one variable equal in both equations, then add or subtract the equations so that variable disappears.

Steps:

1) Choose which variable to remove (eliminate).
2) Multiply each equation by a suitable number so that this variable has the **same coefficient** in both. A simple choice: multiply each equation by the coefficient from the other one.
3) Subtract the equations (or add them, if the coefficients have opposite signs). Equal amounts taken from equal amounts leave equal amounts, so the result is still true, and one variable is gone.
4) Solve for the remaining variable and put its value back in either equation.

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- **Which method to use:**
  - **Substitution** is quick when a variable already has coefficient 1 or −1.
  - **Elimination** is quick when no variable has coefficient 1, or when the coefficients are already equal.
- **Worked examples:**
  - **Example 1:** the incomes of two persons are in the ratio 9 : 7 and their expenditures (money spent) in the ratio 4 : 3.
    - Each saves ₹2000 a month.
    - Find their incomes.
    - Incomes 9x and 7x, expenditures 4y and 3y. Saving = income − expenditure: 9x − 4y = 2000 … (1), 7x − 3y = 2000 … (2).
    - Eliminate (remove) y: (1) × 3 gives 27x − 12y = 6000; (2) × 4 gives 28x − 12y = 8000. Subtracting: **x = 2000**. From (1): 18000 − 4y = 2000, so **y = 4000**.
    - Incomes: **₹18,000** and **₹14,000**. Check: 18000 : 14000 = 9 : 7; expenditures 16000 : 12000 = 4 : 3 ✓.
    - Eliminating x instead: (1) × 7 and (2) × 9 give 63x − 28y = 14000 and 63x − 27y = 18000; subtracting gives y = 4000, the same answer.
  - **Example 2:** the sum of a two-digit number and the number formed by reversing (writing in the opposite order) its digits is 66, and the digits differ by 2. Find the number.
    - Let the digits be x (tens, larger) and y (units). Number = 10x + y; reversed = 10y + x.
    - (10x + y) + (10y + x) = 66 gives 11(x + y) = 66, so x + y = 6; also x − y = 2. Adding: 2x = 8, **x = 4**, **y = 2**.
    - The number is **42** (42 + 24 = 66). If the units digit is the larger one, the number is **24** (24 + 42 = 66). So there are two such numbers, 42 and 24.

✓ **Number of solutions of a pair:**

- A pair of linear equations has exactly one of these: **one solution**, **infinitely many solutions**, or **no solution**.
- **Unique solution:**
  - **Unique solution:** exactly one ordered pair satisfies both equations.
  - During elimination, after one variable is removed, the other one remains.
- **Infinitely many solutions:**
  - **Infinitely many solutions:** one equation is just the other multiplied by a number, so the two are really the same equation.
  - Example: x − y = 10 and 10x − 10y = 100. During elimination, both variables go and we get **0 = 0**, which is always true.
- **No solution:**
  - **No solution:** the left sides can be made the same, but the right sides then differ.
  - Example: x − y = 10 and 10x − 10y = 101. Multiplying the first by 10 gives 10x − 10y = 100, which cannot also be 101. During elimination we get something false, like **0 = 1**.
- **Checking with ratios:**

Table 1: Ratio test for $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$

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Condition	Number of solutions	Graph (two lines)
$a_1/a_2 \neq b_1/b_2$	Unique (one) solution	Intersecting lines
$a_1/a_2 = b_1/b_2 = c_1/c_2$	Infinitely many solutions	Coincident lines
$a_1/a_2 = b_1/b_2 \neq c_1/c_2$	No solution	Parallel lines

- The test also works in reverse: if a pair has a unique solution, the first ratio condition must hold, and so on. The three cases cover every pair, and no pair falls in two cases.
- **When a_2, b_2 or c_2 is zero:** a ratio cannot be formed. Instead, check directly whether one equation is a multiple of the other. Example: x + 2y = 0 and 3x + 6y = 0 (both constants zero) are the same line, as the second is 3 times the first.

Other ways to solve a pair:

- **Graphs:** draw both lines and read where they meet (next topic).
- **Adding and subtracting:** when the x and y coefficients are swapped between the equations, adding and subtracting them gives x + y and x − y at once.
- **Cross-multiplication:** a formula method for any pair.

Worked examples:

- **Example 1:** 10x − 5y = 20 and 4x − 2y = 8: ratios 10/4 = −5/−2 = 20/8 = 5/2.
  - Both are 2x − y = 4, so there are **infinitely many** solutions.
  - More such pairs: x + y = 2 and 3x + 3y = 6; x − 2y = 1 and 2x − 4y = 2; 3x + y = 5 and 6x + 2y = 10.
- **Example 2 (no solution):** x + y = 2 and x + y = 5; 2x − y = 1 and 4x − 2y = 7; x + 3y = 4 and 2x + 6y = 9. In each, the ratios of x and y coefficients are equal but the constants are not in the same ratio.

Where these methods came from:

- Solving systems of linear equations is one of the oldest problems of mathematics. The earliest traces are on **Babylonian cuneiform** (an ancient writing made of wedge-shaped marks on clay) **tablets** (about 2000 BCE).
- In ancient China, the **Nine Chapters on the Mathematical Art** (about 2000 years ago) has a whole chapter, **fangcheng**, on such systems. Coefficients were arranged in columns of counting rods and columns were combined to cancel variables. This is the same as modern elimination, 1800 years before it was named after Gauss in Europe.
- Indian mathematicians **Āryabhaṭa** (499 CE) and **Brahmagupta** (628 CE) also studied systems of equations in several unknowns, using their own methods including elimination.

✓ **Solved questions:**

Form a pair of linear equations and solve.

Q1. The sum of two integers is 5 and their difference is −21. Find them.

➢ x + y = 5 and x − y = −21.

➢ Adding: 2x = −16, x = −8; y = 13.

➢ The integers are **−8 and 13**.

Q2. The difference of two numbers is 26 and one is three times the other. Find them.

➢ x − y = 26 and x = 3y.

➢ So 2y = 26, y = 13, x = 39.

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➢ The numbers are **39 and 13**.

Q3. A coach buys 7 bats and 6 balls for ₹8880, and later 3 bats and 5 balls for ₹4000. Find the cost of a bat and a ball.

➢ $7x + 6y = 8880$ and $3x + 5y = 4000$.

➢ Multiply by 5 and 6: $35x + 30y = 44400$ and $18x + 30y = 24000$.

➢ Subtracting: $17x = 20400$, $x = 1200$.

➢ $3 \times 1200 + 5y = 4000$ gives $y = 80$.

➢ A bat costs **₹1200** and a ball **₹80**.

Q4. Taxi charges are a fixed charge plus a charge per km. For 10 km the fare is ₹155, and for 15 km ₹220. Find the fixed charge and the charge per km. What is the fare for 25 km?

➢ $x + 10y = 155$ and $x + 15y = 220$.

➢ Subtracting: $5y = 65$, $y = 13$; $x = 155 - 130 = 25$.

➢ Fixed charge **₹25**, charge per km **₹13**. For 25 km: $25 + 25 \times 13 =$ **₹350**.

Q5. A fraction becomes 9/11 if 2 is added to both numerator and denominator, and 5/6 if 3 is added to both. Find the fraction.

➢ Let the fraction be x/y.

➢ $(x + 2)/(y + 2) = 9/11$ gives $11x - 9y = -4$; $(x + 3)/(y + 3) = 5/6$ gives $6x - 5y = -3$.

➢ Multiply by 5 and 9: $55x - 45y = -20$ and $54x - 45y = -27$.

➢ Subtracting: $x = 7$; then $9y = 81$, $y = 9$.

➢ The fraction is **7/9**.

Q6. If 1 is added to the numerator and 1 subtracted from the denominator, a fraction becomes 1. It becomes 1/2 if 1 is added only to the denominator. Find the fraction.

➢ $(x + 1)/(y - 1) = 1$ gives $x - y = -2$; $x/(y + 1) = 1/2$ gives $2x - y = 1$.

➢ Subtracting: $x = 3$; $y = 5$.

➢ The fraction is **3/5**.

Q7. Five years ago, Nuri was thrice as old as Sonu. Ten years later, Nuri will be twice as old as Sonu. Find their ages.

➢ Let Nuri be x and Sonu y years. $x - 5 = 3(y - 5)$ gives $x - 3y = -10$; $x + 10 = 2(y + 10)$ gives $x - 2y = 10$.

➢ Subtracting: $y = 20$; $x = 50$.

➢ Nuri is **50 years** and Sonu **20 years** old.

Q8. The sum of the digits of a two-digit number is 9. Nine times the number is twice the number formed by reversing its digits. Find the number.

➢ Tens digit x, units digit y: $x + y = 9$ and $9(10x + y) = 2(10y + x)$, so $88x = 11y$, $y = 8x$.

➢ $x + 8x = 9$ gives $x = 1$, $y = 8$.

➢ The number is **18** ($9 \times 18 = 162 = 2 \times 81$ ✓).

Q9. Meena withdrew ₹2000 in ₹50 and ₹100 notes, 25 notes in all. How many of each?

➢ $x + y = 25$ and $50x + 100y = 2000$ ($x + 2y = 40$)

➢ Subtracting: $y = 15$; $x = 10$.

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- **10 notes of ₹50** and **15 notes of ₹100**.

Q10. A lending library has a fixed charge for the first three days and an extra charge for each day after that. Saritha paid ₹27 for keeping an item for seven days, and Susy paid ₹21 for five days. Find the fixed charge and the charge per extra day.

- Fixed charge ₹x, extra ₹y per day: x + 4y = 27 and x + 2y = 21.
- Subtracting: 2y = 6, y = 3; x = 15.
- Fixed charge **₹15**, extra charge **₹3 per day**.

Q11. Use the ratios a_1/a_2, b_1/b_2, c_1/c_2 to find whether the lines intersect, are parallel or coincident: (i) 5x − 4y + 8 = 0; 7x + 6y − 9 = 0 (ii) 9x + 3y + 12 = 0; 18x + 6y + 24 = 0 (iii) 6x − 3y + 10 = 0; 2x − y + 9 = 0.

- (i) 5/7 ≠ −4/6: **intersecting**.
- (ii) 9/18 = 3/6 = 12/24 = 1/2: **coincident** (lying exactly one upon the other).
- (iii) 6/2 = −3/−1 = 3, but 10/9 ≠ 3: **parallel**.

Q12. Half the perimeter of a rectangular garden, whose length is 4 m more than its width, is 36 m. Find its dimensions.

- l + w = 36 and l = w + 4.
- So 2w + 4 = 36, w = 16, l = 20.
- The garden is **20 m long and 16 m wide**.

Q13. Given 2x + 3y − 8 = 0, write another linear equation so that the pair represents (i) intersecting (ii) parallel (iii) coincident lines.

- (i) **3x − 2y − 7 = 0** (2/3 ≠ 3/−2).
- (ii) **4x + 6y − 5 = 0** (2/4 = 3/6, but −8/−5 ≠ 1/2).
- (iii) **4x + 6y − 16 = 0** (2/4 = 3/6 = −8/−16).

Q14. A problem by Mahāvīrāchārya in Gaṇita sāra saṅgraha (about 850 CE): the price of 9 citrons and 7 fragrant wood-apples is 107; the price of 7 citrons and 9 wood-apples is 101. Find the price of each.

- 9x + 7y = 107 and 7x + 9y = 101.
- The coefficients are **interchanged**, so add and subtract.
- Adding: 16x + 16y = 208, so x + y = 13.
- Subtracting: 2x − 2y = 6, so x − y = 3.
- x = 8, y = 5.
- A citron costs **8** and a wood-apple **5**.

Q15. 5 pencils and 7 pens cost ₹50, while 7 pencils and 5 pens cost ₹46. Find the cost of each.

- 5x + 7y = 50 and 7x + 5y = 46.
- Adding: x + y = 8; subtracting: y − x = 2.
- So x = 3, y = 5.
- A pencil costs **₹3** and a pen **₹5**.

Q16. Find the height of the stool from the figure.

- Let the stool be h cm and the cat c cm high.
- Cat on the stool: h + c = 85.

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- Cat on the floor beside it: $h - c = 25$.
- Adding: $2h = 110$, so the stool is **55 cm** high (and the cat 30 cm).

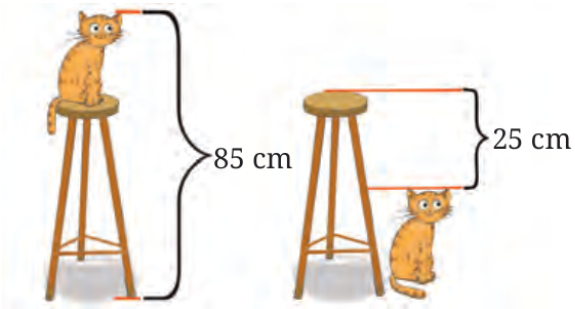


Fig. 18: Cat on the stool (total 85 cm) and cat beside the stool (25 cm below the top)

Q17. For what values of p does the pair $4x + py + 8 = 0$ and $2x + 2y + 2 = 0$ have a unique solution?

- Unique solution when $a_1/a_2 \neq b_1/b_2$: $4/2 \neq p/2$, so $p \neq 4$.
- The pair has a unique solution for **all values of p except 4**.

Q18. Find a and b for which $(a + b)x - 2by = 5a + 2b + 1$ and $3x - y = 14$ have infinitely many solutions.

- Need $(a + b)/3 = -2b/-1 = (5a + 2b + 1)/14$.
- From $(a + b)/3 = 2b$: $a + b = 6b$, so $a = 5b$.
- From $2b = (5a + 2b + 1)/14$: $28b = 25b + 2b + 1$, so **b = 1** and **a = 5**.
- Check: the first equation becomes $6x - 2y = 28$, which is $2 \times (3x - y = 14)$ ✓.

Q19. Find k for which $x + 2y = 3$ and $(k - 1)x + (k + 1)y = k + 3$ are coincident lines.

- Need $1/(k - 1) = 2/(k + 1) = 3/(k + 3)$.
- $1/(k - 1) = 2/(k + 1)$ gives $k + 1 = 2k - 2$, so **k = 3**. Check: $1/2 = 2/4 = 3/6$ ✓.

Q20. At a certain time, Jacob's watch reads 'a' minutes after two o'clock. Fifteen minutes later, it reads 'b' minutes after three o'clock, and $a = 6b$. What time was it at the second look?

- a minutes after 2 plus 15 minutes = b minutes after 3, so $a + 15 = 60 + b$, that is, $a - b = 45$.
- With $a = 6b$: $5b = 45$, $b = 9$, $a = 54$.
- The second time was **3:09** (the first was 2:54).

Q21. The sum of the digits of a two-digit number is 15. The number formed by interchanging the digits is 9 more than the number. Find it.

- $x + y = 15$ and $(10y + x) - (10x + y) = 9$, so $y - x = 1$.
- Then $y = 8$, $x = 7$.
- The number is **78** ($87 - 78 = 9$ ✓).

Q22. In a cyclic quadrilateral ABCD, $\angle A = (x + 7)°$, $\angle B = (y + 8)°$, $\angle C = (3y + 23)°$ and $\angle D = (4x + 12)°$. Find all four angles.

- Opposite angles of a cyclic quadrilateral (a quadrilateral whose four corners lie on a circle) add up to 180°. $\angle A + \angle C = 180$: $x + 3y = 150$. $\angle B + \angle D = 180$: $4x + y = 160$.
- From the second, $y = 160 - 4x$; then $x + 480 - 12x = 150$, so $x = 30$ and $y = 40$.
- **$\angle A = 37°$, $\angle B = 48°$, $\angle C = 143°$, $\angle D = 132°$.**

Q23. A train at uniform speed takes 6 hours less for a distance if its speed is 6 km/h more, and 6 hours more if its speed is 4 km/h less. Find the distance and the speed.

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- Speed v km/h, time t h, distance = vt.
- (v + 6)(t − 6) = vt gives −6v + 6t = 36, so t − v = 6.
- (v − 4)(t + 6) = vt gives 6v − 4t = 24, so 3v − 2t = 12.
- With t = v + 6: 3v − 2v − 12 = 12, so v = 24, t = 30.
- Speed **24 km/h**, distance 24 × 30 = **720 km**.

Q24. A father's age equals the sum of the ages of his four children. After 20 years, the sum of the children's ages will be twice the father's age. Find the father's age.

- Father F, children's total S: F = S. After 20 years, each of the 4 children is 20 years older: S + 80 = 2(F + 20).
- F + 80 = 2F + 40, so the father is **40 years** old.

Q25. Plan A costs ₹50 a month + ₹0.20 per minute; Plan B costs ₹30 a month + ₹0.30 per minute. When is each plan cheaper, and when do they cost the same?

- For t minutes: A = 50 + 0.2t and B = 30 + 0.3t.
- Equal when 0.1t = 20, that is, at **t = 200 minutes** (both cost ₹90).
- **Plan B is cheaper** below 200 minutes; **Plan A is cheaper** above 200 minutes.

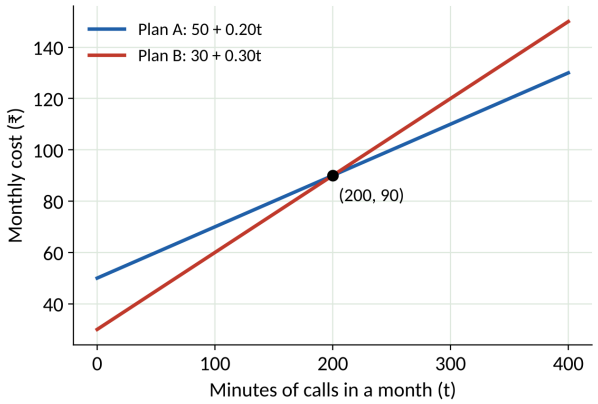


Fig. 19: Monthly cost of the two mobile plans

Graphical Method of Solving a Pair of Linear Equations

✓ **Graphical method:**

- **Graphical method:** draw the lines of both equations on the same axes. A point on **both** lines satisfies both equations, so it is a solution of the pair.
- **Steps:**
  1) For each equation, find at least two points (x = 0 and y = 0 are the easiest).
  2) Plot all the points on the same graph paper.
  3) Draw the two lines and look at how they meet.

✓ **Reading the graph:**

- **Intersecting lines** (cross at one point): that point is the **unique solution**.
- **Parallel lines** (never meet): **no solution**.
- **Coincident lines** (one lies exactly on the other): every point on the line is a solution, so there are **infinitely many solutions**.
- These are the same three cases as in the ratio test (Table 1).

Conceptual Learning

✓ **Lines with equal slope are parallel:**

- Take $y = mx + d_1$ and $y = mx + d_2$, with the same slope m and $d_1 \neq d_2$.
- A common point would need $mx + d_1 = mx + d_2$, that is, $d_1 = d_2$, which is false.
- So the lines have no common point: they are **parallel**. (If $d_1 = d_2$, they are the same line.)

✓ **Worked examples:**

- **Example 1 (intersecting):** solve $x + 3y = 6$ and $2x - 3y = 12$ graphically.
  - Points: (0, 2), (6, 0) on the first; (0, −4), (3, −2) on the second.
  - The lines meet at B (6, 0), so **x = 6, y = 0** is the unique solution.
  - Ratios: $1/2 \neq 3/(-3) = -1$ ✓.
- **Example 2 (parallel):** $x + 2y - 4 = 0$ and $2x + 4y - 12 = 0$: $a_1/a_2 = b_1/b_2 = 1/2$ but $c_1/c_2 = 1/3$.
  - Points R (0, 2), S (4, 0) and P (0, 3), Q (6, 0)
  - The lines never meet: **no solution**.
- **Example 3 (coincident):** Romila buys 2 erasers and 3 A4 sheets for ₹9; Sonali buys 4 erasers and 6 sheets for ₹18. $2x + 3y = 9$ and $4x + 6y = 18$: all ratios are 1/2. Both are the same line through (0, 3), (3, 1), (4.5, 0): **infinitely many solutions** (the prices cannot be found from this data alone).

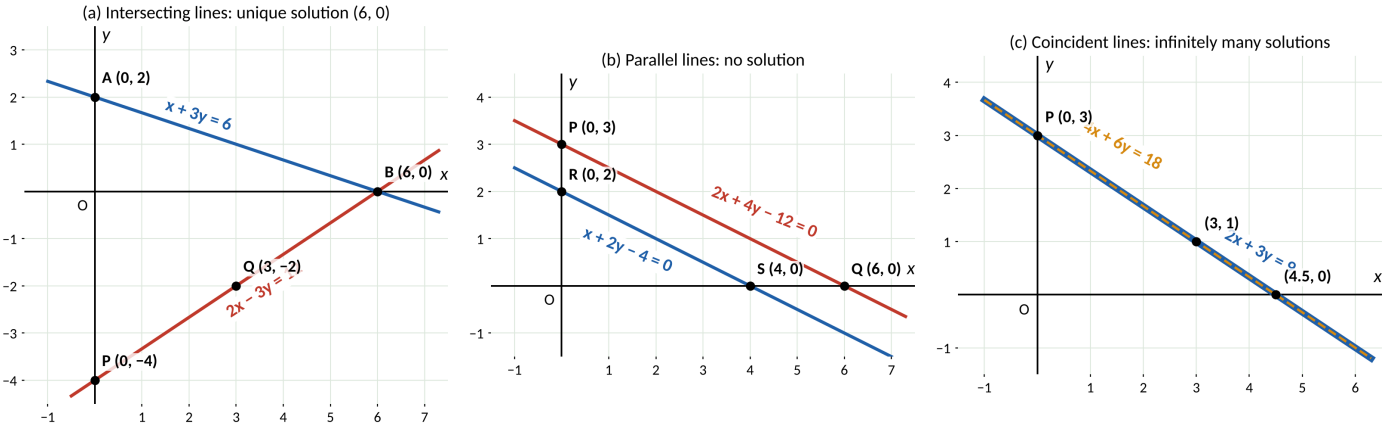


Fig. 20: Graphs of (a) intersecting, (b) parallel and (c) coincident lines

- **Example 4 (four pairs):** drawing the pairs below shows all three cases.
  - Pair 1: $2x + y = 6$ and $x - y = 2$ intersect (cross each other) at **(8/3, 2/3)**: unique solution.
  - Pair 2: $x + y = 4$ and $2x + 2y = 8$, and Pair 3: $3x - 2y = 6$ and $6x - 4y = 12$, are **coincident**: infinitely many solutions.
  - Pair 4: $x - 2y = 4$ and $2x - 4y = 6$ are **parallel**: no solution.

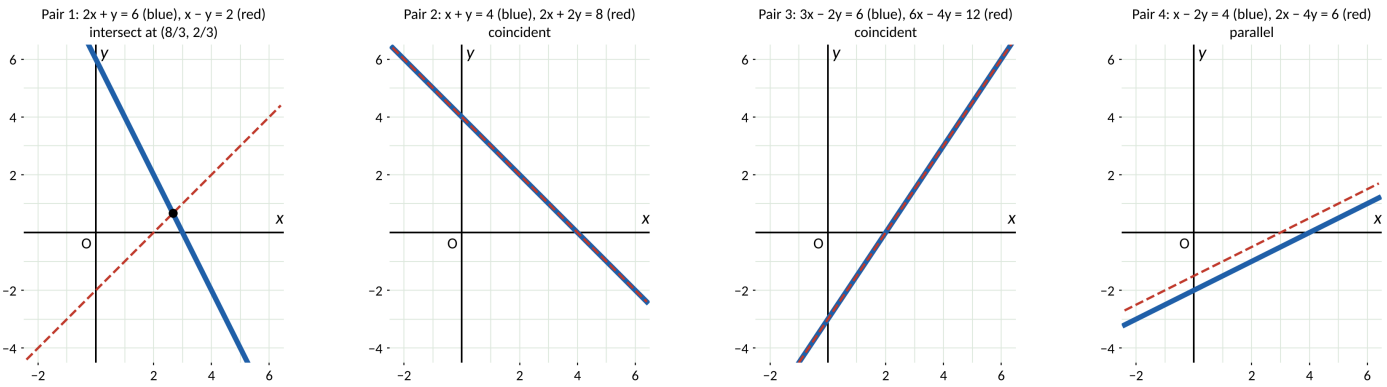


Fig. 21: Graphs of Pairs 1 to 4

✓ **Solved questions:**

Q1. 10 students of Grade 9 took part in a quiz. The number of girls is 4 more than the number of boys. Find the numbers graphically.

- Boys x, girls y: **x + y = 10** and **y = x + 4**.
- Points: (0, 10), (10, 0) for the first; (0, 4), (2, 6) for the second.
- The lines meet at **(3, 7)**: **3 boys and 7 girls**.

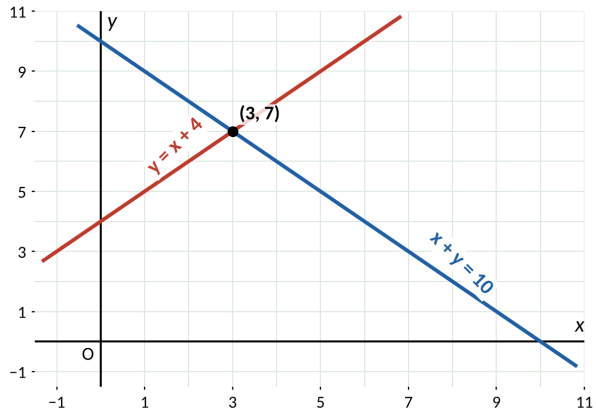


Fig. 22: Graphical solution: 3 boys and 7 girls

Q2. Which pairs have solutions? Find them graphically: (i) x + y = 5, 2x + 2y = 10 (ii) x − y = 8, 3x − 3y = 16 (iii) 2x + y − 6 = 0, 4x − 2y − 4 = 0 (iv) 2x − 2y − 2 = 0, 4x − 4y − 5 = 0.

- (i) 1/2 = 1/2 = 5/10: **coincident**, infinitely many solutions, such as (0, 5), (1, 4), (2, 3).
- (ii) 1/3 = −1/−3 but 8/16 = 1/2: **parallel**, no solution.
- (iii) 2/4 ≠ 1/(−2): **unique solution**. The lines meet at **(2, 2)**.
- (iv) 2/4 = −2/−4 = 1/2 but −2/−5 = 2/5: **parallel**, no solution.

Q3. Solve graphically: 2x + y = 6 and 2x − y − 2 = 0.

- Points: (0, 6), (3, 0) for the first; (0, −2), (1, 0) for the second.
- The lines meet at **(2, 2)**, so x = 2, y = 2.
- (This is the same pair as Q2 (iii).)

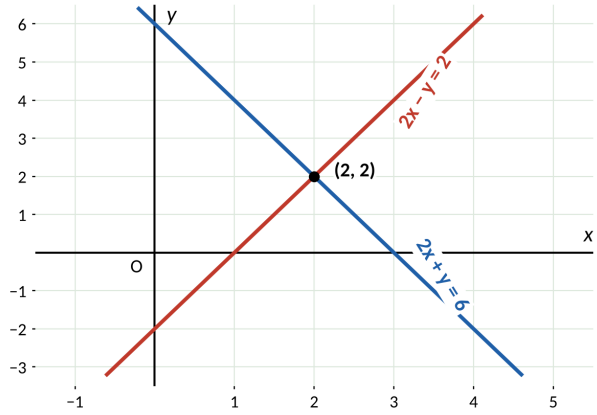


Fig. 23: The lines 2x + y = 6 and 2x − y = 2 meet at (2, 2)

Q4. Find the point of intersection of the lines shown on the cover page.

- The cover page is not available here, so the lines cannot be read.

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- Method: read two points on each line from the cover, write each line's equation (using slope = rise/run and the y-intercept), and solve the pair by any method.

Q5. (i) Robot 1 starts at the origin and repeatedly moves 3 units right and 4 units up. Robot 2 starts at (10, 0) and repeatedly moves 1 unit right and 2 units up. Do their paths meet? (ii) Robot 1 starts at (3, 0) and moves 5 right, 5 up repeatedly; Robot 2 starts at (7, 0) and moves 5 right, 3 down repeatedly. Do their paths meet?

- (i) Robot 1 follows $y = (4/3)x$ (slope 4/3); Robot 2 follows $y = 2(x - 10)$ (slope 2)
- Equating: $4x/3 = 2x - 20$, so $x = 30$, $y = 40$.
- Both robots reach **(30, 40)**: Robot 1 after 10 moves and Robot 2 after 20 moves.
- So their paths **meet at (30, 40)**.
- (ii) Robot 1 follows $y = x - 3$ (going up, $y \geq 0$ for $x \geq 3$); Robot 2 follows $y = -(3/5)(x - 7)$ (going down, $y \leq 0$ for $x \geq 7$)
- The full lines would meet at (4.5, 1.5), but Robot 2 never goes back to $x = 4.5$.
- For $x \geq 7$, Robot 1 is above the x-axis and Robot 2 is below it.
- So the paths **never meet**.

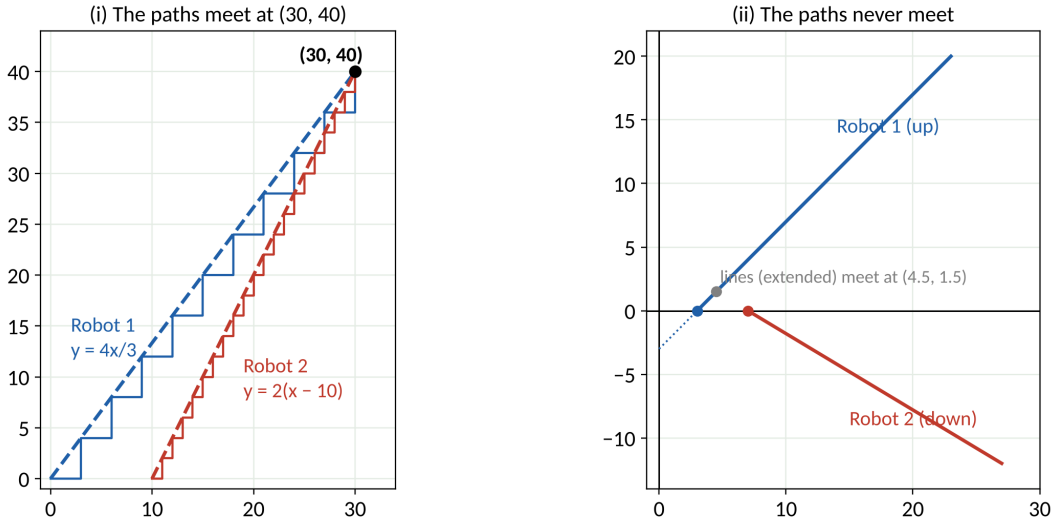


Fig. 24: Paths of the robots in (i) and (ii)

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